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First Author and Second Author

My Institution

Third Author

Other Institution

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Abstract

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Contents

I. POSITIVE EXTENSION CONDITIONS

A. Representability

The necessary and sufficient conditions for the existance of a state that extends a linear functional f_2 defined on $\mathcal{B}(\mathcal{H}_{\mathcal{F}})_{e2}$ to f defined on $\mathcal{B}(\mathcal{H}_{\mathcal{F}})$ are:

1. *Pair Occupation Number Condition (Generalized P Condition)*

$$1 \geq f\left(b_{j_1}^{\dagger} b_{j_2}^{\dagger} b_{j_2} b_{j_1}\right) = f\left(\Phi(\psi_{j_1})^{\dagger} \Phi(\psi_{j_2})^{\dagger} \Phi(\psi_{j_1}) \Phi(\psi_{j_2})\right) \geq 0$$
$$\forall \text{conb's } \{|\psi_l\rangle | 1 \leq l \leq r\} \text{ of } \mathcal{H}^1. \tag{1}$$

   (a) The density operator version of this

$$1 \geq \left(b_{j_1}^{\dagger} b_{j_2}^{\dagger} b_{j_2} b_{j_1} \middle| D\right) = \left(|\psi_{j_1}\psi_{j_2}\rangle \langle \psi_{j_1}\psi_{j_2}|| D_2\right) \geq 0 \tag{2}$$

   (b) The Q-symbol version of this condition is

$$\left\|\tilde{Q}_{D_2}\right\|_{\infty} = \sup_{\mathbf{z}\in\mathbb{C}^{2(r-2)}} \left\{\tilde{Q}_{D_2}\right\} = \sup_{\mathbf{z}\in\mathbb{C}^{r(r-2)}} \left\{\frac{\langle \psi_1(\mathbf{z})\, \psi_2(\mathbf{z})\, | D_2 \psi_1(\mathbf{z})\, \psi_2(\mathbf{z})\rangle}{\det\left(\mathbb{I}_2 + \mathbf{z}^{\dagger}\mathbf{z}\right)}\right\} \leq 1,$$

   if further the extending state describes on the average $\binom{N}{2}$ pairs of fermions then one must add the

2. *Pair Normalization Condition*

$$f(\mathcal{N}_2) = \binom{N}{2}. \tag{3}$$

   a. The density operator version of this is

$$(\mathcal{N}_2 | D) = \left(\sum_{1\leq j_1<j_2\leq r} |\psi_{j_1}\psi_{j_2}\rangle \langle \psi_{j_1}\psi_{j_2}| \middle| D_2\right) = \mathrm{Tr}\{D_2\} = \binom{N}{2}. \tag{4}$$

b. The Q-symbol version of this condition is

$$\int Q_{D_2}(\mathbf{z})\, S(\bar{\mathbf{z}}, \mathbf{z})\, d\mu(\mathbf{z}) = \int \tilde{Q}_{D_2}(\mathbf{z})\, d\tilde{\mu}(\mathbf{z}) = \left\| \tilde{Q}_{D_2} \right\|_1 = \binom{N}{2}. \tag{5}$$

If the linear functional is defined on $\Lambda_2 = \mathcal{B}(\mathcal{H}_{\mathcal{F}})_{e2} \oplus \mathcal{B}(\mathcal{H}_{\mathcal{F}})_{e1}$ then one must consider extra conditions for extending these linear functionals to $\mathcal{B}(\mathcal{H}_{\mathcal{F}})$ and must add the following two conditions:

3. *Generalized G Condition*

$$\begin{aligned} &1 \geq f\left(b_{j_1}^{\dagger} b_{j_2} b_{j_2}^{\dagger} b_{j_1}\right) = f\left(\Phi(\psi_{j_1})^{\dagger} \Phi(\psi_{j_2}) \Phi(\psi_{j_2})^{\dagger} \Phi(\psi_{j_1})\right) \geq 0 \\ &\forall \text{conb's } \{|\psi_l\rangle | 1 \leq l \leq r\} \text{ of } \mathcal{H}^1, \end{aligned} \tag{6}$$

which is equivalent to

$$\begin{aligned} 1 &\geq f\left(b_{j_1}^{\dagger} b_{j_1}\right) - f\left(b_{j_1}^{\dagger} b_{j_2}^{\dagger} b_{j_2} b_{j_1}\right) \geq 0, \\ 1 &\geq f\left(b_{j_2}^{\dagger} b_{j_2}\right) - f\left(b_{j_1}^{\dagger} b_{j_2}^{\dagger} b_{j_2} b_{j_1}\right) \geq 0 \end{aligned} \tag{7}$$

and

$$\begin{aligned} 1 &\geq f\left(\Phi(\psi_{j_1})^{\dagger} \Phi(\psi_{j_1})\right) - f\left(\Phi(\psi_{j_1})^{\dagger} \Phi(\psi_{j_2})^{\dagger} \Phi(\psi_{j_2}) \Phi(\psi_{j_1})\right) \geq 0, \\ 1 &\geq f\left(\Phi(\psi_{j_2})^{\dagger} \Phi(\psi_{j_2})\right) - f\left(\Phi(\psi_{j_1})^{\dagger} \Phi(\psi_{j_2})^{\dagger} \Phi(\psi_{j_2}) \Phi(\psi_{j_1})\right) \geq 0, \end{aligned} \tag{8}$$

a. the density operator version of this is

$$\begin{aligned} 1 &\geq (|\psi_{j_1}\rangle \langle \psi_{j_1}|| D_1) - (|\psi_{j_1} \psi_{j_2}\rangle \langle \psi_{j_1} \psi_{j_2}|| D_2) \geq 0 \\ 1 &\geq (|\psi_{j_2}\rangle \langle \psi_{j_2}|| D_1) - (|\psi_{j_1} \psi_{j_2}\rangle \langle \psi_{j_1} \psi_{j_2}|| D_2) \geq 0, \end{aligned} \tag{9}$$

b. the Q-symbol version of this condition is

$$\begin{aligned} 1 &\geq \left(\left|\tilde{\psi}_1(\mathbf{z})\right\rangle \left\langle \tilde{\psi}_1(\mathbf{z})\right|\Big| D_1\right) - \left(\left|\tilde{\psi}_1(\mathbf{z}) \tilde{\psi}_2(\mathbf{z})\right\rangle \left\langle \tilde{\psi}_1(\mathbf{z}) \tilde{\psi}_2(\mathbf{z})\right|\Big| D_2\right) \geq 0 \\ 1 &\geq \left(\left|\tilde{\psi}_2(\mathbf{z})\right\rangle \left\langle \tilde{\psi}_2(\mathbf{z})\right|\Big| D_1\right) - \left(\left|\tilde{\psi}_1(\mathbf{z}) \tilde{\psi}_2(\mathbf{z})\right\rangle \left\langle \tilde{\psi}_1(\mathbf{z}) \tilde{\psi}_2(\mathbf{z})\right|\Big| D_2\right) \geq 0, \end{aligned} \tag{10}$$

which can be expressed as

$$1 \geq Q_{1D_1} \geq Q_{D_2} \geq 0 \tag{11}$$

$$1 \geq Q_{2D_1} \geq Q_{D_2} \geq 0 \tag{12}$$

and the

4. *Generalized Q Condition* is

$$
\begin{aligned}
&1 \geq f\left(b_{j_2} b_{j_1} b_{j_1}^{\dagger} b_{j_2}^{\dagger}\right) \geq 0 \Leftrightarrow 1 \geq 1-f\left(b_{j_2}^{\dagger} b_{j_2}\right)-f\left(b_{j_1}^{\dagger} b_{j_1}\right)+f\left(b_{j_1}^{\dagger} b_{j_2}^{\dagger} b_{j_2} b_{j_1}\right) \geq 0 \\
&\Leftrightarrow 0 \geq f\left(b_{j_1}^{\dagger} b_{j_2}^{\dagger} b_{j_2} b_{j_1}\right)-f\left(b_{j_2}^{\dagger} b_{j_2}\right)-f\left(b_{j_1}^{\dagger} b_{j_1}\right) \geq-1 \\
&\Leftrightarrow 1 \geq f\left(b_{j_2}^{\dagger} b_{j_2}\right)+f\left(b_{j_1}^{\dagger} b_{j_1}\right)-f\left(b_{j_1}^{\dagger} b_{j_2}^{\dagger} b_{j_2} b_{j_1}\right) \geq 0
\end{aligned} \tag{13}
$$

equivalently

$$
1 \geq f\left(\Phi\left(\psi_{j_2}\right)^{\dagger} \Phi\left(\psi_{j_2}\right)\right)+f\left(\Phi\left(\psi_{j_2}\right)^{\dagger} \Phi\left(\psi_{j_2}\right)\right)-f\left(\Phi\left(\psi_{j_1}\right)^{\dagger} \Phi\left(\psi_{j_2}\right)^{\dagger} \Phi\left(\psi_{j_2}\right) \Phi\left(\psi_{j_1}\right)\right) \geq 0. \tag{14}
$$

This condition is a result of the fact that one has the idempotent identities

$$
I_{\mathcal{H}_{\mathcal{F}}}-b_{j_2} b_{j_1} b_{j_1}^{\dagger} b_{j_2}^{\dagger}=b_{j_1}^{\dagger} b_{j_1}+b_{j_2}^{\dagger} b_{j_2}-b_{j_1}^{\dagger} b_{j_2}^{\dagger} b_{j_2} b_{j_1} \tag{15}
$$

$$
\left(I_{\mathcal{H}_{\mathcal{F}}}-b_{j_2} b_{j_1} b_{j_1}^{\dagger} b_{j_2}^{\dagger}\right)^2=I_{\mathcal{H}_{\mathcal{F}}}-b_{j_2} b_{j_1} b_{j_1}^{\dagger} b_{j_2}^{\dagger}. \tag{16}
$$

a. The density operator version of this is

$$
1 \geq\left(\left|\psi_{j_2}\right\rangle\left\langle\psi_{j_2}\right| \| D_1\right)+\left(\left|\psi_{j_1} \psi_{j_2}\right\rangle\left\langle\psi_{j_1} \psi_{j_2}\right| \| D_2\right)-\left(\left|\psi_{j_1} \psi_{j_2}\right\rangle\left\langle\psi_{j_1} \psi_{j_2}\right| \| D_2\right) \geq 0. \tag{17}
$$

b. The Q-symbol version of this condition is

$$
1 \geq\left(\left|\tilde{\psi}_1(\mathbf{z})\right\rangle\left\langle\tilde{\psi}_1(\mathbf{z})\right| \| D_1\right)+\left(\left|\tilde{\psi}_2(\mathbf{z})\right\rangle\left\langle\tilde{\psi}_2(\mathbf{z})\right| \| D_1\right)-\left(\left|\tilde{\psi}_1(\mathbf{z}) \tilde{\psi}_2(\mathbf{z})\right\rangle\left\langle\tilde{\psi}_1(\mathbf{z}) \tilde{\psi}_2(\mathbf{z})\right| \| D_2\right) \geq 0.
$$

5. *Occupation Number Condition*

$$
\begin{gathered}
1 \geq f\left(b_j^{\dagger} b_j\right)=f\left(\Phi\left(\psi_j\right)^{\dagger} \Phi\left(\psi_j\right)\right) \geq 0 \\
\forall \text{conb's } \left\{\left|\psi_l\right\rangle \mid 1 \leq l \leq r\right\} \text{ of } \mathcal{H}^1
\end{gathered} \tag{18}
$$

a. The density operator version of this is

$$
1 \geq\left(\left|\psi_j\right\rangle\left\langle\psi_j\right| \| D_1\right) \geq 0 \tag{19}
$$

b. The Q-symbol version of this condition is

$$
\begin{aligned}
&1 \geq\left(\left|\tilde{\psi}_1(\mathbf{z})\right\rangle\left\langle\tilde{\psi}_1(\mathbf{z})\right| \| D_1\right) \geq 0 \\
&1 \geq\left(\left|\tilde{\psi}_2(\mathbf{z})\right\rangle\left\langle\tilde{\psi}_2(\mathbf{z})\right| \| D_1\right) \geq 0
\end{aligned} \tag{20}
$$

6. *Particle Normalization Condition*

$$f\left(\mathcal{N}_1\right)=N. \tag{21}$$

a. The density operator version of this is

$$\left(\mathcal{N}_1\right|D)=\left(\sum_{1\le j_1\le r}\left|\psi_{j_1}\right\rangle\left\langle\psi_{j_1}\right|\left|D_1\right.\right)=\operatorname{Tr}\left\{D_1\right\}=N. \tag{22}$$

b. The Q-symbol version of this condition is

$$\int\left(\left\{\left|\tilde{\psi}_1(\mathbf{z})\right\rangle\left\langle\tilde{\psi}_1(\mathbf{z})\right|+\left|\tilde{\psi}_2(\mathbf{z})\right\rangle\left\langle\tilde{\psi}_2(\mathbf{z})\right|\right\}\right|D_1\Big)\,d\tilde{\mu}(\mathbf{z})=N. \tag{23}$$

The set of operators satisfying the condition (1) - (6) form the set $\mathcal{S}_2$ of Second Order Reduced states i.e. the set of Representable SORDO's.

B. N-Representability

In order for a linear functional be the restriciton of a pure number state it must satisfy

$$f_2\left(\mathcal{N}_1^2\right)=f_2\left(\mathcal{N}_1\right)^2=N^2 \tag{24}$$

i.e. its expectation value must be dispersion free on the number operator, where N is some fixed integer. Thus f_2 must be defined on Λ_1, D_1 a linear functional of D_2 (a necessary condition for f_2 to be extended to a pure number state) and the values of f_2 on $\mathcal{B}\left(\mathcal{H}_{\mathcal{F}}\right)_{e1}$ related to the values on $\mathcal{B}\left(\mathcal{H}_{\mathcal{F}}\right)_{e2}$ by $f_2\left(b_j^{\dagger}b_k\right)=(N-1)^{-1}f_2\left(b_j^{\dagger}\mathcal{N}_1b_k\right)$. If the

1 *Pair Normalization Condition*

$$f\left(\mathcal{N}_2\right)=\binom{N}{2} \tag{25}$$

a. The density operator version of this is

$$\left(\mathcal{N}_2\right|D)=\left(\sum_{1\le j_1<j_2\le r}\left|\psi_{j_1}\psi_{j_2}\right\rangle\left\langle\psi_{j_1}\psi_{j_2}\right|\left|D_2\right.\right)=\operatorname{Tr}\left\{D_2\right\}=\binom{N}{2}. \tag{26}$$

b. The Q-symbol version of this condition is

$$\int Q_{D_2}(\mathbf{z})\,S\left(\bar{\mathbf{z}},\mathbf{z}\right)d\mu(\mathbf{z})=\int\tilde{Q}_{D_2}(\mathbf{z})\,d\tilde{\mu}(\mathbf{z})=\left\|\tilde{Q}_{D_2}\right\|_1=\binom{N}{2}. \tag{27}$$

is satisfied then eq. (**??**) is equivalent to

$$f_2\left(\mathcal{N}_1^2\right) = f_2\left(\mathcal{N}_1\right)^2 \tag{28a}$$

and

$$f_2\left(\mathcal{N}_1\right) = N, \tag{29}$$

i.e. f_2 is dispersion free on the number operator.

The necessary and sufficient conditions for the existance of a pure number state f that is the extension of a linear functional f_2 defined on $\mathcal{B}\left(\mathcal{H}_{\mathcal{F}}\right)_{e2}$ (and extended to $\mathcal{B}\left(\mathcal{H}_{\mathcal{F}}\right)_{e1}$in the preceding manner) to $\mathcal{B}\left(\mathcal{H}_{\mathcal{F}}\right)$ are:

2. *Pair Occupation Number Condition (Generalized P Condition)*

$$\begin{aligned} &1 \geq f\left(b_{j_1}^{\dagger} b_{j_2}^{\dagger} b_{j_2} b_{j_1}\right) = f\left(\Phi\left(\psi_{j_1}\right)^{\dagger} \Phi\left(\psi_{j_2}\right)^{\dagger} \Phi\left(\psi_{j_1}\right) \Phi\left(\psi_{j_2}\right)\right) \geq 0 \\ &\forall \text{conb's } \left\{\left|\psi_l\right\rangle \middle| 1 \leq l \leq r\right\} \text{ of } \mathcal{H}^1. \end{aligned} \tag{30}$$

a. The density operator version of this

$$1 \geq \left(b_{j_1}^{\dagger} b_{j_2}^{\dagger} b_{j_2} b_{j_1} \middle| D\right) = \left(\left|\psi_{j_1}\psi_{j_2}\right\rangle \left\langle \psi_{j_1}\psi_{j_2}\right| \middle| D_2\right) \geq 0 \tag{31}$$

b. The Q-symbol version of this condition is

$$\left\|\tilde{Q}_{D_2}\right\|_{\infty} = \sup_{\mathbf{z}\in\mathbb{C}^{2(r-2)}} \left\{\tilde{Q}_{D_2}\right\} = \sup_{\mathbf{z}\in\mathbb{C}^{r(r-2)}} \left\{\frac{\left\langle \psi_1\left(\mathbf{z}\right)\psi_2\left(\mathbf{z}\right)\middle| D_2 \psi_1\left(\mathbf{z}\right)\psi_2\left(\mathbf{z}\right)\right\rangle}{\det\left(\mathbb{I}_2 + \mathbf{z}^{\dagger}\mathbf{z}\right)}\right\} \leq 1,$$

1. *Generalized G Condition*

$$\begin{aligned} &1 \geq f\left(b_{j_1}^{\dagger} b_{j_2} b_{j_2}^{\dagger} b_{j_1}\right) = f\left(\Phi\left(\psi_{j_1}\right)^{\dagger} \Phi\left(\psi_{j_2}\right) \Phi\left(\psi_{j_2}\right)^{\dagger} \Phi\left(\psi_{j_1}\right)\right) \geq 0 \\ &\forall \text{conb's } \left\{\left|\psi_l\right\rangle \middle| 1 \leq l \leq r\right\} \text{ of } \mathcal{H}^1, \end{aligned} \tag{32}$$

which is equivalent to

$$\begin{aligned} 1 &\geq f\left(b_{j_1}^{\dagger} b_{j_1}\right) - f\left(b_{j_1}^{\dagger} b_{j_2}^{\dagger} b_{j_2} b_{j_1}\right) \geq 0, \\ 1 &\geq f\left(b_{j_2}^{\dagger} b_{j_2}\right) - f\left(b_{j_1}^{\dagger} b_{j_2}^{\dagger} b_{j_2} b_{j_1}\right) \geq 0 \end{aligned} \tag{33}$$

and

$$\begin{aligned} 1 &\geq f\left(\Phi\left(\psi_{j_1}\right)^{\dagger} \Phi\left(\psi_{j_1}\right)\right) - f\left(\Phi\left(\psi_{j_1}\right)^{\dagger} \Phi\left(\psi_{j_2}\right)^{\dagger} \Phi\left(\psi_{j_2}\right) \Phi\left(\psi_{j_1}\right)\right) \geq 0, \\ 1 &\geq f\left(\Phi\left(\psi_{j_2}\right)^{\dagger} \Phi\left(\psi_{j_2}\right)\right) - f\left(\Phi\left(\psi_{j_1}\right)^{\dagger} \Phi\left(\psi_{j_2}\right)^{\dagger} \Phi\left(\psi_{j_2}\right) \Phi\left(\psi_{j_1}\right)\right) \geq 0, \end{aligned} \tag{34}$$

a. the density operator version of this is

$$1 \geq \left(|\psi_{j_1}\rangle \langle\psi_{j_1}| \right| D_1) - \left(|\psi_{j_1}\psi_{j_2}\rangle \langle\psi_{j_1}\psi_{j_2}| \right| D_2) \geq 0$$
$$1 \geq \left(|\psi_{j_2}\rangle \langle\psi_{j_2}| \right| D_1) - \left(|\psi_{j_1}\psi_{j_2}\rangle \langle\psi_{j_1}\psi_{j_2}| \right| D_2) \geq 0, \tag{35}$$

b. the Q-symbol version of this condition is

$$1 \geq \left(\left|\tilde{\psi}_1(\mathbf{z})\right\rangle \left\langle \tilde{\psi}_1(\mathbf{z})\right|\right| D_1\Big) - \left(\left|\tilde{\psi}_1(\mathbf{z})\tilde{\psi}_2(\mathbf{z})\right\rangle \left\langle \tilde{\psi}_1(\mathbf{z})\tilde{\psi}_2(\mathbf{z})\right|\right| D_2\Big) \geq 0$$
$$1 \geq \left(\left|\tilde{\psi}_2(\mathbf{z})\right\rangle \left\langle \tilde{\psi}_2(\mathbf{z})\right|\right| D_1\Big) - \left(\left|\tilde{\psi}_1(\mathbf{z})\tilde{\psi}_2(\mathbf{z})\right\rangle \left\langle \tilde{\psi}_1(\mathbf{z})\tilde{\psi}_2(\mathbf{z})\right|\right| D_2\Big) \geq 0, \tag{36}$$

which can be expressed as

$$1 \geq \left[\mathcal{L}_{11}\left(\tilde{Q}_{D_2}\right)\right](\mathbf{z}) - \tilde{Q}_{D_2}(\mathbf{z}) \geq 0 \tag{37}$$

$$1 \geq \left[\mathcal{L}_{22}\left(\tilde{Q}_{D_2}\right)\right](\mathbf{z}) - \tilde{Q}_{D_2}(\mathbf{z}) \geq 0 \tag{38}$$

i.e.

$$1 \geq \mathcal{L}_{11}\left(\tilde{Q}_{D_2}\right) - \tilde{Q}_{D_2} \geq 0 \tag{39}$$

$$1 \geq \mathcal{L}_{22}\left(\tilde{Q}_{D_2}\right) - \tilde{Q}_{D_2} \geq 0 \tag{40}$$

2. *Generalized Q Condition* is

$$1 \geq f\left(b_{j_2}b_{j_1}b_{j_1}^\dagger b_{j_2}^\dagger\right) \geq 0 \Leftrightarrow 1 \geq 1 - f\left(b_{j_2}^\dagger b_{j_2}\right) - f\left(b_{j_1}^\dagger b_{j_1}\right) + f\left(b_{j_1}^\dagger b_{j_2}^\dagger b_{j_2}b_{j_1}\right) \geq 0$$
$$\Leftrightarrow 0 \geq f\left(b_{j_1}^\dagger b_{j_2}^\dagger b_{j_2}b_{j_1}\right) - f\left(b_{j_2}^\dagger b_{j_2}\right) - f\left(b_{j_1}^\dagger b_{j_1}\right) \geq -1$$
$$\Leftrightarrow 1 \geq f\left(b_{j_2}^\dagger b_{j_2}\right) + f\left(b_{j_1}^\dagger b_{j_1}\right) - f\left(b_{j_1}^\dagger b_{j_2}^\dagger b_{j_2}b_{j_1}\right) \geq 0 \tag{41}$$

equivalently

$$1 \geq f\left(\Phi(\psi_{j_2})^\dagger \Phi(\psi_{j_2})\right) + f\left(\Phi(\psi_{j_2})^\dagger \Phi(\psi_{j_2})\right) - f\left(\Phi(\psi_{j_1})^\dagger \Phi(\psi_{j_2})^\dagger \Phi(\psi_{j_2})\Phi(\psi_{j_1})\right) \geq 0. \tag{42}$$

This condition is a result of the fact that one has the idempotent identities

$$I_{\mathcal{H}_\mathcal{F}} - b_{j_2}b_{j_1}b_{j_1}^\dagger b_{j_2}^\dagger = b_{j_1}^\dagger b_{j_1} + b_{j_2}^\dagger b_{j_2} - b_{j_1}^\dagger b_{j_2}^\dagger b_{j_2}b_{j_1} \tag{43}$$

$$\left(I_{\mathcal{H}_\mathcal{F}} - b_{j_2}b_{j_1}b_{j_1}^\dagger b_{j_2}^\dagger\right)^2 = I_{\mathcal{H}_\mathcal{F}} - b_{j_2}b_{j_1}b_{j_1}^\dagger b_{j_2}^\dagger. \tag{44}$$

a. The density operator version of this is

$$1 \geq \left(\left| \psi_{j_2} \right\rangle \left\langle \psi_{j_2} \right| \right| D_1) + \left(\left| \psi_{j_1} \psi_{j_2} \right\rangle \left\langle \psi_{j_1} \psi_{j_2} \right| \right| D_2) - \left(\left| \psi_{j_1} \psi_{j_2} \right\rangle \left\langle \psi_{j_1} \psi_{j_2} \right| \right| D_2) \geq 0. \tag{45}$$

b. The Q-symbol version of this condition is

$$1 \geq \left(\left| \tilde{\psi}_1 (\mathbf{z}) \right\rangle \left\langle \tilde{\psi}_1 (\mathbf{z}) \right| \right| D_1 \Big) + \left(\left| \tilde{\psi}_2 (\mathbf{z}) \right\rangle \left\langle \tilde{\psi}_2 (\mathbf{z}) \right| \right| D_1 \Big) - \left(\left| \tilde{\psi}_1 (\mathbf{z}) \tilde{\psi}_2 (\mathbf{z}) \right\rangle \left\langle \tilde{\psi}_1 (\mathbf{z}) \tilde{\psi}_2 (\mathbf{z}) \right| \right| D_2 \Big) \geq 0,$$

i.e.

$$1 \geq \tilde{Q}_{\Gamma_1^2 \circ L_2^1 (D_2)} (\mathbf{z}) - \tilde{Q}_{D_2} (\mathbf{z}) = \left[\mathcal{L}_{\Gamma L} \left(\tilde{Q}_{D_2} \right) \right] (\mathbf{z}) - \tilde{Q}_{D_2} (\mathbf{z}) \geq 0 \tag{46}$$

and

$$1 \geq \mathcal{L}_{\Gamma L} \left(\tilde{Q}_{D_2} \right) - \tilde{Q}_{D_2} \geq 0 \tag{47}$$

3. *Occupation Number Condition*

$$\begin{gathered} 1 \geq f \left(b_j^\dagger b_j \right) = f \left(\Phi (\psi_j)^\dagger \Phi (\psi_j) \right) \geq 0 \\ \forall \text{conb's } \{ |\psi_l\rangle | \, 1 \leq l \leq r \} \text{ of } \mathcal{H}^1 \end{gathered} \tag{48}$$

The density operator version of this is

$$1 \geq \left(|\psi_j\rangle \langle \psi_j | \right| D_1) \geq 0 \tag{49}$$

The Q-symbol version of this condition is

$$\begin{gathered} 1 \geq \left(\left| \tilde{\psi}_1 (\mathbf{z}) \right\rangle \left\langle \tilde{\psi}_1 (\mathbf{z}) \right| \right| D_1 \Big) \geq 0 \\ 1 \geq \left(\left| \tilde{\psi}_2 (\mathbf{z}) \right\rangle \left\langle \tilde{\psi}_2 (\mathbf{z}) \right| \right| D_1 \Big) \geq 0 \end{gathered} \tag{50}$$

i.e.

$$1 \geq \left[\mathcal{L}_{11} \left(\tilde{Q}_{D_2} \right) \right] (\mathbf{z}) \geq 0 \tag{51}$$

$$1 \geq \left[\mathcal{L}_{22} \left(\tilde{Q}_{D_2} \right) \right] (\mathbf{z}) \geq 0. \tag{52}$$

and

$$1 \geq \mathcal{L}_{11} \left(\tilde{Q}_{D_2} \right) \geq 0 \tag{53}$$

$$1 \geq \mathcal{L}_{22} \left(\tilde{Q}_{D_2} \right) \geq 0. \tag{54}$$

The functionals $\mathcal{L}_{11}, \mathcal{L}_{22}$ and $\mathcal{L}_{\Gamma L}$ are defined by

$$\begin{aligned}\mathcal{L}_{jk}\left(\mathbf{z},\mathbf{z}'\right) &= \frac{1}{2\left(N-1\right)}\sum_{1\leq j,k,l\leq r} P_{jlkl}\left(\mathbf{z}'\right)\mathrm{Tr}\left\{\left|\tilde{\varphi}_1\left(\mathbf{z}\right)\right\rangle\left\langle\tilde{\varphi}_1\left(\mathbf{z}\right)\right|\left|\varphi_j\right\rangle\left\langle\varphi_k\right|\right\} K_Q\left(\mathbf{z},\mathbf{z}'\right)\\ &= \frac{1}{2\left(N-1\right)}\sum_{1\leq j,k,l\leq r}\gamma_1\left(\mathbf{z}\right)_{kj} K_Q\left(\mathbf{z},\mathbf{z}'\right) P_{jlkl}\left(\mathbf{z}'\right).\end{aligned} \tag{55}$$

$$\left[\mathcal{L}_{\Gamma L}\left(Q_{D_2}\right)\right]\left(\mathbf{z}\right) = Q_{\Gamma_1^2\circ L_1(D)}\left(\mathbf{z}\right) = \left(\left|\varphi_1\left(\mathbf{z}\right)\varphi_2\left(\mathbf{z}\right)\right\rangle\left\langle\varphi_1\left(\mathbf{z}\right)\varphi_2\left(\mathbf{z}\right)\right|\Big|\,\Gamma_1^2\circ L_1\left(D\right)\right)_{\mathcal{B}(\mathcal{H}^2)} \tag{56}$$

We denote the convex set of representable Q-symbols as $\mathfrak{Q}$, the convex set of N-representable Q-symbols as $\mathfrak{Q}_N$ and the unit ball wrt the supremum (uniform) norm of the Q-symbol space, $\mathfrak{F}$, as $\mathfrak{Q}_1$, i.e.

$$\mathfrak{Q}_1 = \left\{Q\left(\mathbf{z}\right)\left|\left|Q\left(\mathbf{z}\right)\right|\leq 1, \mathbf{z}\in\mathbb{C}^{2(r-2)}\right.\right\}, \tag{57}$$

which is actually the unit hypercube in $\mathfrak{F}$. The extreme points, $\mathrm{ext}\left\{\mathfrak{Q}_1\right\}$, of this hypercube are the vertices of the surface

$$S^{\mathbb{C}^{2(r-2)}} = \left\{Q\left(\mathbf{z}\right)\left|\left|Q\left(\mathbf{z}\right)\right| = 1, \mathbf{z}\in\mathbb{C}^{2(r-2)}\right.\right\}, \tag{58}$$

the intersection with the positive hyperant is

$$\mathfrak{Q}_{1+} = \left\{Q\left(\mathbf{z}\right)\left|0\leq Q\left(\mathbf{z}\right)\leq 1, \mathbf{z}\in\mathbb{C}^{2(r-2)}\right.\right\} \tag{59}$$

and the positive segment of the hypersphere is

$$S_+^{\mathbb{C}^{2(r-2)}} = \left\{Q\left(\mathbf{z}\right)\left|Q\left(\mathbf{z}\right) = 1, \mathbf{z}\in\mathbb{C}^{2(r-2)}\right.\right\}. \tag{60}$$

C. Energy Functional

In terms of P and Q symbols the N-fermion energy functional can be expressed as

$$E_N\left(D^2\right) = \int Q_{D^2}\left(\mathbf{z}\right)P_{K_{2N}}\left(\mathbf{z}\right)S\left(\bar{\mathbf{z}},\mathbf{z}\right)d\mu\left(\mathbf{z}\right) = \int Q_{K_{2N}}\left(\mathbf{z}\right)P_{D^2}\left(\mathbf{z}\right)S\left(\bar{\mathbf{z}},\mathbf{z}\right)d\mu\left(\mathbf{z}\right), \tag{61}$$

which comes from the general trace product expression for self adjoint trace class operators

$$\mathrm{Tr}\left\{AB\right\} = \int Q_B\left(\mathbf{z}\right)P_A\left(\mathbf{z}\right)S\left(\bar{\mathbf{z}},\mathbf{z}\right)d\mu\left(\mathbf{z}\right) = \int Q_A\left(\mathbf{z}\right)P_B\left(\mathbf{z}\right)S\left(\bar{\mathbf{z}},\mathbf{z}\right)d\mu\left(\mathbf{z}\right).$$

Again this can be expressed as

$$\operatorname{Tr}\{AB\} = \int Q_B(\mathbf{z})\, P_A(\mathbf{z})\, d\tilde{\mu}(\mathbf{z}) = \int Q_A(\mathbf{z})\, P_B(\mathbf{z})\, d\tilde{\mu}(\mathbf{z}) \tag{62}$$

and

$$E_N\left(D^2\right) = \int Q_{D^2}(\mathbf{z})\, P_{K_{2N}}(\mathbf{z})\, d\tilde{\mu}(\mathbf{z}) = \int Q_{K_{2N}}(\mathbf{z})\, P_{D^2}(\mathbf{z})\, d\tilde{\mu}(\mathbf{z})\,. \tag{63}$$

$$E_N\left(D^2\right) = \int Q_{D^2}(\mathbf{z})\, P_{K_{2N}}(\mathbf{z})\, d\tilde{\mu}(\mathbf{z}) = \int Q_{K_{2N}}(\mathbf{z})\, P_{D^2}(\mathbf{z})\, d\tilde{\mu}(\mathbf{z})\,. \tag{64}$$

The Q-symbol of the Reduced Hamiltonian is given by

$$\begin{aligned} Q_{K_{2N}}(\mathbf{z}) &= \frac{1}{4} \sum_{1 \le j_1, j_2, k_1, k_2 \le r} K_{2N j_1 j_2 k_1 k_2} \left\langle \mathbf{z} \left| a_{j_1}^{\dagger} a_{j_2}^{\dagger} a_{k_2} a_{k_1} \mathbf{z} \right. \right\rangle \\ &= \frac{1}{4} \langle \mathbf{z} \,|\mathbf{z} \rangle \sum_{1 \le j_1, j_2, k_1, k_2 \le r} K_{2N j_1 j_2 k_1 k_2} \left\{ \gamma_1(\mathbf{z})_{j_1 k_1} \gamma_1(\mathbf{z})_{j_2 k_2} - \gamma_1(\mathbf{z})_{j_1 k_2} \gamma_1(\mathbf{z})_{j_2 k_1} \right\}, \end{aligned} \tag{65}$$

where

$$\begin{aligned} K_{2N j_1 j_2 k_1 k_2} &= \frac{2}{N(N-1)} \eta \left(\delta_{j_1 k_1} \delta_{j_2 k_2} - \delta_{j_1 k_2} \delta_{j_2 k_1}\right) \\ &\quad + \frac{1}{(N-1)} \left(h_{1 j_1 k_1} \delta_{j_2 k_2} + h_{1 j_2 k_2} \delta_{j_1 k_1}\right) + \frac{1}{2} h_{2 j_1 j_2 k_1 k_2}. \end{aligned} \tag{66}$$

Hence

$$\tilde{Q}_{K_{2N}}(\mathbf{z}) = E_{NIPS}(\gamma_1(\mathbf{z})) = \eta + \operatorname{Tr}_{\mathcal{H}^1} \left\{ h_1 \gamma_1(\mathbf{z}) - \frac{1}{2} V_{HF}(\mathbf{z}) \right\}, \tag{67}$$

where the Hartree-Fock potential V_{HF} is given as

$$V_{HF}(\mathbf{z})_{jk} = \sum_{1 \le l, m \le r} h_{2N jlkm} \gamma_1(\mathbf{z})_{lm}\,. \tag{68}$$

The P-symbol $\tilde{P}_{K_{2N}}(\mathbf{z})$ of $\tilde{Q}_{K_{2N}}$ is then given as

$$\tilde{P}_{K_{2N}}(\mathbf{z}) = \left[\mathfrak{Be}^{-1}\left(\tilde{Q}_{K_{2N}}\right)\right](\mathbf{z}) \tag{69}$$

and the N-particle energy functional becomes the linear functional

$$E_N\left(D^2\right) = E_N\left(Q_{D^2}\right) = \int Q_{D^2}(\mathbf{z})\, \tilde{P}_{K_{2N}}(\mathbf{z})\, S(\bar{\mathbf{z}}, \mathbf{z})\, d\mu(\mathbf{z})\,. \tag{70}$$

The ground state Q-symbol $E_N\left(Q_{D^2_{GS}}\right)$ energy its energy E_{NGS} and the is then solution to the Linear Programing problem

$$E_N\left(Q_{D^2_{GS}}\right) = \min_{Q_{D^2} \in \mathfrak{Q}_N} E_N\left(Q_{D^2}\right) \tag{71}$$

where $\mathfrak{Q}_N$ is the set determined by the preceding linear constraints.

II. APPENDIX

The expectation value of the number operator is then

$$\begin{aligned} f\left(\mathcal{N}_1\right) &= (N-1)^{-1}\sum_j f\left(a_j^\dagger \mathcal{N}_1 a_j\right) = (N-1)^{-1}\sum_j f\left(\left(\mathcal{N}_1 a_j^\dagger - a_j^\dagger\right)a_j\right) \\ &= (N-1)^{-1} f\left(\mathcal{N}_1^2\right) - (N-1)^{-1} f\left(\mathcal{N}_1\right) \end{aligned}$$

using

$$\left[\mathcal{N}_1, a_j^\dagger\right] = a_j^\dagger = \mathcal{N}_1 a_j^\dagger - a_j^\dagger \mathcal{N}_1 \Leftrightarrow a_j^\dagger \mathcal{N}_1 = \mathcal{N}_1 a_j^\dagger - a_j^\dagger \tag{72}$$

one obtains

$$\begin{aligned} f\left(\mathcal{N}_1\right) &= (N-1)^{-1} f\left(\mathcal{N}_1^2\right) - (N-1)^{-1} f\left(\mathcal{N}_1\right) \\ &\Rightarrow (N-1) f\left(\mathcal{N}_1\right) = f\left(\mathcal{N}_1^2\right) - f\left(\mathcal{N}_1\right) \\ &\Rightarrow N f\left(\mathcal{N}_1\right) = f\left(\mathcal{N}_1^2\right) \end{aligned} \tag{73}$$

then using

$$\left(\mathcal{N}_1\right)^2 = \mathcal{N}_1 + 2\mathcal{N}_2 \tag{74}$$

one gets

$$\begin{aligned} N f\left(\mathcal{N}_1\right) &= f\left(\mathcal{N}_1 + 2\mathcal{N}_2\right) = f\left(\mathcal{N}_1\right) + 2 f\left(\mathcal{N}_2\right) \\ &\Rightarrow \frac{(N-1)}{2} f\left(\mathcal{N}_1\right) = f\left(\mathcal{N}_2\right) \end{aligned} \tag{75}$$

thus if

$$f\left(\mathcal{N}_2\right) = \frac{N(N-1)}{2} \tag{76}$$

then

$$f\left(\mathcal{N}_1\right) = N \tag{77}$$

and

$$f\left(\mathcal{N}_1^2\right) = N^2 = f\left(\mathcal{N}_1\right)^2 . \tag{78}$$

$$\begin{aligned} \mathcal{L}_{11}\left(\mathbf{z}, \mathbf{z}'\right) &= \frac{1}{2(N-1)} \sum_{1\le j_1,k_1,l\le r} P_{j_1 l k_1 l}\left(\mathbf{z}'\right) \operatorname{Tr}\left\{\left|\tilde{\varphi}_1(\mathbf{z})\right\rangle \left\langle \tilde{\varphi}_1(\mathbf{z})\right| \left|\varphi_{j_1}\right\rangle \left\langle \varphi_{k_1}\right|\right\} K_Q\left(\mathbf{z}, \mathbf{z}'\right) \\ &= \frac{1}{2(N-1)} \sum_{1\le j_1,k_1,l\le r} \gamma_1(\mathbf{z})_{k_1 j_1} K_Q\left(\mathbf{z}, \mathbf{z}'\right) P_{j_1 l k_1 l}\left(\mathbf{z}'\right) . \end{aligned} \tag{79}$$

$$Q_{\Gamma_1^2\circ L_1(D)}(\mathbf{z}) = \left(|\varphi_1(\mathbf{z})\,\varphi_2(\mathbf{z})\rangle\,\langle\varphi_1(\mathbf{z})\,\varphi_2(\mathbf{z})| \middle| \Gamma_1^2 \circ L_1(D)\right)_{\mathcal{B}(\mathcal{H}^2)} \tag{80}$$

The SORDM for *any* ensemble, D, of pure number states is defined as

$$D_{2j_1j_2k_1k_2} = \left(a_{j_1}^\dagger a_{j_2}^\dagger a_{k_2} a_{k_1} \middle| D\right)_{\mathcal{F}} = \mathrm{Tr}\left\{ a_{k_1}^\dagger a_{k_2}^\dagger a_{j_2} a_{j_1} D\right\} = \sum_{0\le N\le\infty} \mathrm{Tr}\left\{ a_{k_1}^\dagger a_{k_2}^\dagger a_{j_2} a_{j_1} D^N\right\}$$
$$= \sum_{0\le N\le\infty} \mathrm{Tr}\left\{ a_{k_1}^\dagger a_{k_2}^\dagger \mathcal{N}_{N-2} a_{j_2} a_{k_2} D^N\right\} = \sum_{\substack{0\le N\le\infty\\ 1\le l_1<\cdots<l_{N-2}\le\infty}} \mathrm{Tr}\left\{ a_{k_1}^\dagger a_{k_2}^\dagger a_{l_1}^\dagger \cdots a_{l_{N-2}}^\dagger a_{l_{N-2}} \cdots a_{l_1} a_{j_2} a_{j_1} D^N\right\}$$
$$= \sum_{\substack{0\le N\le\infty\\ 1\le l_1<\cdots<l_{N-2}\le\infty}} D^N_{j_1j_2l_1\cdots l_{N-2}k_1k_2l_1\cdots l_{N-2}} = L_2(D)_{j_1j_2k_1k_2}\,, \tag{81}$$

which gives

$$D_{2j_1j_2k_1k_2} = \sum_{0\le N\le\infty} \binom{N}{2} L_N^2\left(D^N\right)_{j_1j_2k_1k_2} = \sum_{0\le N\le\infty} L_2\left(D^N\right)_{j_1j_2k_1k_2}. \tag{82}$$

and the FORDM by

$$D_{1j_1k_1} = \left(a_{j_1}^\dagger a_{k_1} \middle| D\right)_{\mathcal{F}} = \mathrm{Tr}\left\{ a_{k_1}^\dagger a_{j_1} D\right\} = \sum_{0\le N\le\infty} \mathrm{Tr}\left\{ a_{k_1}^\dagger a_{j_1} D^N\right\}$$
$$= \sum_{0\le N\le\infty} \mathrm{Tr}\left\{ a_{k_1}^\dagger \mathcal{N}_{N-1} a_{j_1} D^N\right\} = \sum_{\substack{0\le N\le\infty\\ 1\le l_1<\cdots<l_{N-2}\le\infty}} \mathrm{Tr}\left\{ a_{k_1}^\dagger a_{l_1}^\dagger \cdots a_{l_{N-1}}^\dagger a_{l_{N-1}} \cdots a_{l_1} a_{j_1} D^N\right\}$$
$$= \sum_{\substack{0\le N\le\infty\\ 1\le l_1<\cdots<l_{N-1}\le\infty}} D^N_{j_1j_2l_1\cdots l_{N-1}k_1k_2l_1\cdots l_{N-1}} = L_1(D)_{j_1k_1}\,, \tag{83}$$

which gives

$$D_{1j_1k_1} = \sum_{0\le N\le\infty} \binom{N}{1} L_N^1\left(D^N\right)_{j_1k_1} = \sum_{0\le N\le\infty} L_1\left(D^N\right)_{j_1k_1}. \tag{84}$$

Hence if $D = D^N$ for a fixed N then

$$D_{2j_1j_2k_1k_2} = \sum_{1\le l_1<\cdots<l_{N-2}\le\infty} D^N_{j_1j_2l_1\cdots l_{N-2}k_1k_2l_1\cdots l_{N-2}} \tag{85}$$

and

$$D_{1j_1k_1} = \sum_{1\le l_1<\cdots<l_{N-1}\le\infty} D^N_{j_1j_2l_1\cdots l_{N-1}k_1k_2l_1\cdots l_{N-1}} \tag{86}$$

and

$$D_{1j_1k_1} = \frac{1}{2(N-1)} \sum_{1\le l\le\infty} D_{2j_1lk_1l}. \tag{87}$$

Theorem 1 *f is a pure N-particle state iff*

$$f\left(a_j^\dagger a_k\right) = (N-1)^{-1} f\left(a_j^\dagger \mathcal{N}_1 a_k\right); \quad 1 \le j,k \le r \tag{88}$$

for any basis.

Proof. Proof. If f is a pure N particle state eq. (88) follows immediately and the "any basis" assertion follows from the invariance of $\mathcal{N}_1$ to $U\left(\mathcal{H}^1\right)$.

Let f be a mixed particle state i.e.

$$f(X) = \sum_{0\le P\le\infty} \alpha_P \left\langle \Psi_P \left| X \Psi_P \right.\right\rangle; 1 \ge \alpha_P \ge 0; \sum_{0\le P\le\infty} \alpha_P = 1 \tag{89}$$

then

$$\begin{aligned} f\left(a_j^\dagger \mathcal{N}_1 a_k\right) &= \sum_{0\le P\le\infty} \alpha_P \left\langle \Psi_P \left| a_j^\dagger \mathcal{N}_1 a_k \Psi_P \right.\right\rangle = \sum_{0\le P\le\infty} \alpha_P (P-1) \left\langle \Psi_P \left| a_j^\dagger a_k \Psi_P \right.\right\rangle \\ &= (N-1) \sum_{0\le P\le\infty} \alpha_P \left\langle \Psi_P \left| a_j^\dagger a_k \Psi_P \right.\right\rangle \end{aligned} \tag{90}$$

and thus

$$\begin{aligned} 0 &= \sum_{0\le P\le\infty} \alpha_P \{P-N\} \left\langle \Psi_P \left| a_j^\dagger a_k \Psi_P \right.\right\rangle \forall j,k \\ &\Rightarrow \sum_{0\le P\le\infty} \alpha_P \{P-N\} \left\langle \Psi_P \left| b^\dagger b \Psi_P \right.\right\rangle = 0 \forall b \\ &\Leftrightarrow \sum_{0\le P\le\infty} \alpha_P \{P-N\} n_b = 0; \quad \forall n_b \in [0,1] \\ &\Leftrightarrow \sum_{0\le P\le\infty} \alpha_P \{P-N\} = 0 = \sum_{0\le P\le\infty} \alpha_P P - N \\ &\Leftrightarrow \sum_{0\le P\le\infty} \alpha_P P = N; \quad P \in \mathbb{N}, \sum_{0\le P\le\infty} \alpha_P = 1, \alpha_P \in [0,1] \\ &\Leftrightarrow \alpha_N = 1, \alpha_P = 0, P \ne N. \end{aligned} \tag{91}$$

■ ■

$$\begin{aligned} \left[\mathcal{N}_1, a_j^\dagger\right] &= a_j^\dagger = \mathcal{N}_1 a_j^\dagger - a_j^\dagger \mathcal{N}_1 \\ \left[\mathcal{N}_1, a_j\right] &= -a_j = a_j \mathcal{N}_1 - \mathcal{N}_1 a_j \Leftrightarrow \mathcal{N}_1 a_j = a_j + a_j \mathcal{N}_1 \end{aligned} \tag{92}$$